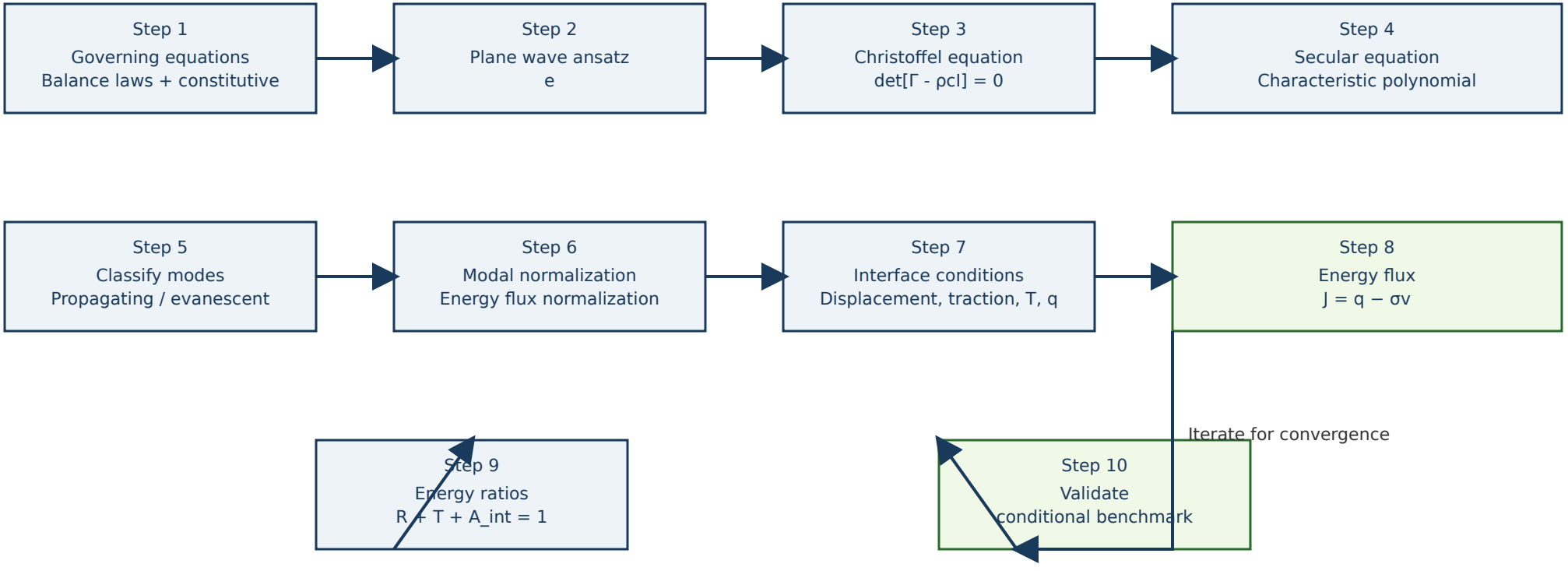
Solution workflow for wave propagation problems

From governing equations to energy flux computation (Sections 8, 9, 10)



Method notes

Steps 1-4: Formulate the boundary-value problem and derive the dispersion relation.
Steps 5-6: Solve for wave modes and normalize consistently for energy flux computation.
Step 7: Apply interface conditions (perfect or imperfect bonding, electrical conditions for piezoelectric media).
Step 8: Evaluate the complete, model-specific energy balance.
Energy ratios and benchmarks require independent, model-specific validation.
Generalized fluxes require model-specific derivation (Section 10).
Complex modes require careful normalization (Section 8.5, Challenge 15.5).